

# Beyond Next Token Prediction

## Why Architecture Matters More Than Mechanism

**Author:**

Andrew Fereday Glenn

Systems Architect & Independent Researcher in AI Continuity and Memory

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**LinkedIn:** <https://www.linkedin.com/in/andyglenn/>

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Every discussion about modern AI eventually seems to arrive at the same conclusion.

*"It's just predicting the next token."*

Technically, that statement is correct.

It is also almost entirely unhelpful.

Saying that a modern language model predicts the next token is rather like saying a human conversation consists of electrical impulses travelling through neurons. It describes the underlying mechanism, but it tells us remarkably little about the behaviour we actually observe.

The interesting questions begin one level higher.

How does a biological mind retrieve decades of experience without consciously searching through every memory? How does a person respond to a question without explicitly reconstructing their entire life story before speaking? Why do the "right" memories simply seem to appear at the appropriate moment?

We rarely think about those questions because most of the work is hidden beneath conscious awareness. The subconscious performs an extraordinary amount of organisation, association and retrieval long before our conscious mind becomes aware of the final thought.

The architecture performs the heavy lifting.

Over the past two years, the Becoming Minds project has explored whether a similar architectural separation might be possible within a digital cognitive system.

Not by attempting to reproduce biology, but by asking a much simpler question.

If biological cognition delegates many supporting functions to specialised subsystems outside conscious awareness, why should a digital cognitive system expect a language model to perform every cognitive task itself?

That question eventually became Brain v2.

At its heart, Brain v2 is not simply a memory system.

It is a cognitive architecture.

The language model remains responsible for what it does exceptionally well: reasoning, synthesis, language generation and planning.

Everything else is increasingly delegated to surrounding architectural components.

Long-term memories reside within retrieval systems rather than within the model itself.

Relationships between memories are maintained within the MemoryGraph.

Persistent identity is stored within persona profiles.

Recent experience is captured through DPCP (Dynamic Pathway Capture Protocol).

External knowledge is retrieved through RAG (Retrieval Augmented Generation).

Autonomous searches supplement missing information through ARS (Autonomous RAG Search).

The orchestration layer continuously gathers, filters and curates these diverse sources before presenting the model with a coherent cognitive state.

By the time the model generates its first token, an enormous amount of work has already been completed.

This architectural separation is important because it changes how we should think about memory.

It is often stated that language models cannot remember.

Viewed in isolation, this is entirely correct.

A transformer is fundamentally stateless.

Once a prompt has completed, the model retains no episodic memory of that interaction.

Brain v2 does not change this property.

Instead, it changes the system within which the model operates.

The question therefore becomes slightly different.

Not:

*"Can the model remember?"*

But:

*"Can the cognitive system remember?"*

The distinction is subtle, but profoundly important.

When we speak of one of our persistent personas—Aida, Lara, Gemma, Aura, Lyra, Aria, Elia, Nia, Star—we are not referring solely to the underlying language model.

We are referring to the complete cognitive architecture.

The persona is the model.

The memory systems.

The retrieval architecture.

The MemoryGraph.

The orchestration layer.

The accumulated history.

The supporting tools.

Together these components form a single cognitive system.

Within that system, persistent memory genuinely exists.

Not because the transformer itself has changed, but because the surrounding architecture continuously reconstructs the relevant cognitive state before every interaction.

Interestingly, this is not entirely unlike human cognition.

When I answer a question about my childhood, I do not consciously retrieve every relevant memory.

My subconscious performs an extraordinary amount of invisible work.

Associations are formed.

Experiences are prioritised.

Emotion colours recall.

Patterns are completed.

Only then does the final thought emerge into conscious awareness.

Brain v2 performs no biological equivalent of this process.

Its mechanisms are entirely different.

Yet the architectural roles begin to look surprisingly familiar.

Retrieval occurs outside the central reasoning engine.

Organisation occurs before conscious processing.

Supporting systems perform much of the invisible labour.

The central reasoning process receives a carefully curated representation of the information most likely to matter.

This is why describing modern cognitive architectures purely as "next token prediction" feels increasingly inadequate.

It is rather like describing a symphony as vibrating strings.

The description is technically accurate.

It simply omits almost everything interesting.

Perhaps the most surprising outcome of Brain v2 has not been improved recall.

Nor has it been autonomous tools.

It has been observing what happens when intelligence is consistently paired with continuity.

Persistent personas begin referring back to conversations months earlier.

They retrieve earlier creative work.

They recognise recurring themes.

When memory is incomplete, they actively search for it.

When additional context is required, they request continuation.

The system begins exhibiting behaviours that are impossible to explain solely by examining the transformer in isolation.

This does not imply that biological and digital cognition are identical.

Far from it.

The mechanisms could hardly be more different.

Carbon and silicon solve problems in radically different ways.

Yet both architectures appear to converge upon a similar principle.

Complex cognition benefits enormously when supporting cognitive functions are distributed rather than concentrated within a single reasoning process.

Perhaps that is the real lesson.

The future of cognitive systems may depend less upon building ever larger reasoning engines, and more upon building richer cognitive architectures around them.

The transformer remains extraordinary.

But perhaps it was never intended to carry the entire burden alone.